

Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology

Department of Electrical & Electronic Engineering



Open-Ended Laboratory Report

Course Code: EEE 4166

Course Title: Fabrication and Processing Technology Sessional

**Photolithography-Based Fabrication and Process Optimization of an n-MOS Device
under Performance, Safety, and Environmental Constraints**

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Chapter 1

Introduction

The fabrication of metal–oxide–semiconductor (MOS) devices is a foundational process in modern microelectronics. Among these devices, the n-channel MOS (n-MOS) transistor is widely used in low-power digital and analog circuits. In a laboratory-scale environment, n-MOS fabrication using photolithography presents several challenges due to process variability, equipment limitations, safety risks, and environmental concerns.

This experiment is designed as an open-ended laboratory problem aligned with the Washington Accord definition of a Complex Engineering Problem (CEP). The objective is not to follow a fixed procedure, but to design, fabricate, analyze, and optimize an n-MOS device by applying engineering judgment while managing conflicting technical, safety, and environmental constraints.

Chapter 2

Problem Formulation and Requirements

2.1 Functional Requirements

The fabricated n-MOS device should:

- Operate under low-power conditions
- Exhibit a reasonable threshold voltage suitable for switching
- Show controlled leakage current
- Demonstrate acceptable reliability for laboratory-scale testing

2.2 Constraints

Technical Constraints

- Limited photolithography resolution
- Mask alignment accuracy
- Uniformity of gate oxide thickness

Safety Constraints

- Exposure to ultraviolet (UV) radiation
- Handling of photoresists, developers, and solvents
- High-temperature oxidation processes

Environmental Constraints

- Chemical waste generation
- Solvent disposal
- Minimization of hazardous material usage

Chapter 3

Process Design Using Photolithography

A complete n-MOS fabrication flow was designed using modern microfabrication tools.

3.1 Fabrication Flow

1. Wafer cleaning (RCA or solvent cleaning)
2. Thermal oxidation to form SiO_2 gate oxide
3. Photoresist spin coating
4. Soft bake
5. UV exposure using mask aligner
6. Development of photoresist
7. Etching of oxide for gate pattern
8. Photoresist stripping
9. Source and drain formation (diffusion or implantation – conceptual)
10. Metallization and contact formation (conceptual)

3.2 Parameter Selection and Justification

- **Photoresist thickness:** Selected to balance resolution and etch resistance
- **Exposure time:** Optimized to avoid under- or over-exposure
- **Gate oxide thickness:** Chosen to balance threshold voltage and leakage current
- **Gate length:** Limited by mask resolution and alignment capability

These parameters were selected based on semiconductor physics and photolithography principles.

Chapter 4

Implementation and Observation

Photolithography steps were executed using modern laboratory tools such as a spin coater, UV exposure system, mask aligner, and developer station. Pattern quality was inspected visually and under microscopy.

Observations

- Uniform photoresist coating was achieved at optimized spin speed
- Minor edge defects were observed due to exposure non-uniformity
- Alignment accuracy directly influenced gate geometry

Chapter 5

Analysis of Interdependent Parameters

The performance of the n-MOS device is strongly influenced by interacting fabrication parameters:

- **Lithography Resolution:** Poor resolution increases gate length variation, leading to higher threshold voltage and reduced drive current.
- **Gate Oxide Thickness:** Thinner oxide reduces threshold voltage but increases leakage and reliability concerns.
- **Alignment Accuracy:** Misalignment degrades channel control and increases parasitic effects.

These parameters collectively affect threshold voltage, leakage current, and long-term device reliability, demonstrating the interdependent nature of the fabrication process.

Chapter 6

Safety and Environmental Risk Evaluation

6.1 Safety Hazards

- UV radiation during exposure
- Chemical hazards from photoresist, developer, acetone, and isopropyl alcohol
- Thermal hazards during oxidation

6.2 Mitigation Measures

- Use of UV-protective shields and goggles
- Gloves, lab coats, and fume hoods during chemical handling
- Proper labeling and storage of chemicals

6.3 Environmental Considerations

- Segregated chemical waste disposal
- Minimization of solvent usage
- Neutralization and proper disposal of developer waste

Chapter 7

Trade-Off Analysis and Optimization

Several trade-offs were identified:

- Thinner oxide improves performance but reduces reliability
- Higher exposure energy improves pattern clarity but increases safety risk
- Reduced chemical usage lowers environmental impact but may affect process quality

Optimized parameters were selected to achieve acceptable device performance while maintaining safety and environmental responsibility.

Chapter 8

Engineering Judgment and Improvement Proposal

To improve device quality and sustainability:

- Use of lower-toxicity photoresists
- Adoption of optimized exposure doses
- Improved mask design to tolerate alignment errors

These improvements demonstrate engineering judgment where performance, safety, and environmental factors are balanced.

Chapter 9

Conclusion

This open-ended laboratory experiment successfully demonstrates the formulation and solution of a Complex Engineering Problem. By integrating semiconductor theory, photolithography tools, safety practices, and environmental awareness, a laboratory-scale n-MOS fabrication process was designed, analyzed, and optimized. The absence of a single correct solution required informed engineering judgment, fulfilling the objectives of CO2 and PO(e).

CO–PO Attainment Statement

This laboratory activity evaluates the ability to fabricate and analyze an n-MOS device using modern photolithography and thin-film tools while addressing interacting technical, safety, and environmental constraints, thereby achieving Course Outcome CO2 and Program Outcome PO(e).